

Digital Mental Health Interventions for Prevention and Treatment of Social Anxiety Disorder in Children, Adolescents and Young Adults: Systematic Review and Meta-Analysis of Randomised Controlled Trials.

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Digital Mental Health Interventions for Prevention and Treatment of Social Anxiety Disorder in Children, Adolescents and Young Adults: Systematic Review and Meta-Analysis of Randomised Controlled Trials.

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Abstract

Background: Social anxiety disorder (SAD) significantly affects young individuals' social and academic functioning, emphasising the need for accessible and effective treatments such as digital mental health interventions (DMHI).

Objective: This systematic review and meta-analysis aimed to evaluate the efficacy of DMHI for children, adolescents and young adults with social anxiety.

Methods: For review and meta-analysis databases were searched for randomised controlled trials investigating DMHI addressing social anxiety in young people (mean age <25 years). Two authors independently screened the records, extracted data and assessed the Risk of Bias.

Results: This systematic review includes k=22 and the meta-analysis k=21 studies. Results significantly favoured DMHIs ($g=0.508$; 95% CI: 0.308; 0.707) over any control condition (i.e., waitlist or active interventions) after the intervention, specifically those compared to waitlist control conditions ($g=0.576$; 95% CI: 0.343; 0.809), based on cognitive behavioural principles ($g=0.610$; 95% CI: 0.361; 0.859), incorporating SAD-specific components ($g=0.878$; 95% CI: 0.469; 1.278) and those with human guidance ($g=0.825$; 95% CI: 0.425; 1.224). Neither parental involvement nor age influenced outcomes significantly. When publication bias was considered, the overall effect remained significant ($g=0.506$ (95% CI= 0.308; 0.707)).

Conclusions: This meta-analysis provides support for the efficacy of DMHI for social anxiety compared to no intervention and the beneficial effects of guidance and interventions specifically designed for SAD. Further, it highlights methodological shortcomings and heterogeneous reporting standards. Future research should prioritise higher methodological quality and should explore how effects are related to age and specific intervention components, including guidance and treatment modules. Clinical Trial: This systematic review and meta-analysis was preregistered on PROSPERO (ID: CRD42023424181).

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Original Paper

Digital Mental Health Interventions for Prevention and Treatment of Social Anxiety Disorder in Children, Adolescents and Young Adults: Systematic Review and Meta-Analysis of Randomised Controlled Trials.

Abstract

Background: Social anxiety disorder (SAD) significantly affects young individuals' social and academic functioning, emphasising the need for accessible and effective treatments such as digital mental health interventions (DMHI).

Objective: This systematic review and meta-analysis aimed to evaluate the efficacy of DMHI for children, adolescents and young adults with social anxiety.

Methods: For review and meta-analysis six electronic databases (APA PsycINFO, Embase, MEDLINE, PsycIndex, PubMed, Web of Science) were searched for randomised controlled trials investigating DMHI addressing social anxiety in young people (mean age <25 years). Two authors independently screened the records, extracted data and assessed the Risk of Bias. For data analysis, a standardised effect size was calculated using Hedges' g , along with 95% confidence intervals for each study. Meta-analyses were conducted using a random-effects model to account for heterogeneity.

Results: This systematic review includes $k=22$ and the meta-analysis $k=21$ studies. Results significantly favoured DMHIs ($g=0.508$; 95% CI: 0.308; 0.707) over any control condition (i.e., waitlist or active interventions) after the intervention, specifically those compared to waitlist control conditions ($g=0.576$; 95% CI: 0.343; 0.809), based on cognitive behavioural principles ($g=0.610$; 95% CI: 0.361; 0.859), incorporating SAD-specific components ($g=0.878$; 95% CI: 0.469; 1.278) and those with human guidance ($g=0.825$; 95% CI: 0.425; 1.224). Neither parental involvement nor age influenced outcomes significantly. When publication bias was considered, the overall effect remained significant ($g=0.506$; 95% CI= 0.308; 0.707). Risk of bias assessment indicated that most studies showed some concerns, three studies were classified as high risk and three were rated as low risk. The reporting of adherence varied greatly and could not be analysed meta-analytically.

Conclusion: This meta-analysis provides support for the efficacy of DMHI for social anxiety compared to control conditions and the beneficial effects of guidance and interventions specifically designed for SAD. Further, it highlights methodological shortcomings and heterogeneous reporting standards. Future research should prioritise higher methodological quality and should explore how effects are related to age and specific intervention components, including guidance and treatment modules.

Trial Registration: This systematic review and meta-analysis was preregistered on PROSPERO (ID: CRD42023424181).

Keywords: Social anxiety; digital mental health intervention; digital mental health prevention; children; adolescents; young adults; meta-analysis; systematic review



Introduction

Social Anxiety Disorder in Children, Adolescents and Young Adults

Anxiety disorders are the most prevalent mental disorders in children, adolescents and young adults [1–3]. Especially in adolescence, social anxiety disorder (SAD) is highly relevant, with prevalence rates between 12 to 36% [4,5]. Notably, 88% of SAD cases have their onset before the age of 25 years and 50% before 14 years [6]. The first symptoms (e.g., shyness, fear of embarrassment) often appear in childhood [7,8] and in adolescence up to 50% report subclinical levels of social anxiety symptoms [9,10].

SAD is characterised by an intense and persistent fear of embarrassing oneself in front of others or being judged negatively by others [11]. To endure feared situations, those affected by SAD often engage in safety behaviours to minimise negative evaluations or avoid social situations altogether [11]. These behaviours maintain social anxiety symptoms [12–14], contribute to substantial impairment in their psychosocial functioning (e.g., fewer friends than their healthy peers) and the development of comorbid disorders (e.g., depression, substance use) and lead to negative long-term effects, such as poorer academic and job opportunities and lack of social support due to a small or non-existent social network [15–20]. These detrimental consequences demand prevention efforts and effective treatments to intervene early.

Efficacy of Treatment and Barriers to Treatment

Established treatment protocols (e.g., based on the Cognitive Model by Clark and Wells [12]) in adults have been adapted to suit the specific needs of children and adolescents [21]. Meta-analyses support their efficacy in individual and group settings compared to waitlist or psychological placebo control groups (i.e., attention control) [22–24]. Accordingly, individual or group cognitive behavioural therapy (CBT) is recommended as a first-line treatment for children and adolescents with SAD [25,26].

Although effective treatments exist, treatment delivery and uptake are impeded by poor mental health literacy (e.g., poor symptom recognition), public- and self-stigma, particularly in those with SAD, and a gap in the provision of care [27–30]. The care gap is even more exacerbated in children and adolescents, in ethnic minority groups, those living in rural areas or in lower-income countries and those with lower socio-economic status [31–33]. Various efforts have addressed some of these barriers by integrating care in primary physician settings, implementing psychoeducational interventions in educational settings, or by task sharing (e.g., non-professional people deliver interventions under supervision) [34–36]. Digital interventions are a highly scalable and far-reaching solution that represents another promising option that may circumvent some of the barriers not addressed in the previously mentioned efforts (i.e., time and place independence).

Digital Mental Health Interventions for Social Anxiety

Digital mental health interventions (DMHI) aim to manage, alleviate, or treat mental health problems through a digital medium (i.e., internet, mobile phone application (app), wearables, SMS) [37,38]. They can include some form of human support (i.e., guided self-help), for instance, through asynchronous messages or synchronous real-time chatting or phone calls by a professional, or they can be completely standalone (i.e., unguided self-help) [39]. Several DMHIs have been developed for adult SAD, with the evidence base for SAD being among the most extensive, including interventions based on CBT, psychodynamic therapy, interpersonal therapy and acceptance-based CBT [39,40]. For this age group, meta-analytic evidence supports the efficacy of DMHIs as internet-based CBT (ICBT) and virtual reality exposure significantly outperformed passive or waitlist control conditions. Compared to other active control conditions, ICBT yielded small effects and showed to be equally efficacious as face-to-face psychotherapy [41–43].

In children and adolescents, meta-analyses on the efficacy of DMHIs have focused predominantly on

depressive and/or anxiety symptoms but have not reported effect sizes for either the prevention or treatment of SAD specifically [37,38]. Although SAD was displayed separately in one meta-analysis, no pooled effect size was calculated [44]. For DMHIs for anxiety disorders in general, meta-analyses reported significant effects for interventions based on CBT principles compared to a waitlist control condition (ages 6 to 16: $SMD = 0.68$; ages 7 to 18: $g = 1.41$; ages 6 to 25: $g = 0.68$) and no significant difference to other active conditions (e.g., face-to-face: $g = 0.30$; $SMD = -0.04$) [45–47]. For DMHIs for the prevention of anxiety disorders, meta-analyses did not yield significant overall effects, but some RCTs found small effects in favour of DMHIs [48,49]. Likewise, meta-analyses of prevention programmes for anxiety disorders in school-based settings reported small to medium effect sizes compared to passive control conditions after the intervention [50,51].

Notably, some studies suggest that young people with SAD benefit less from transdiagnostic interventions that target several anxiety disorders than young people with other anxiety disorders [52–54]. However other studies found no difference between SAD-specific and transdiagnostic anxiety interventions for children and adolescents with SAD [55,56]. Thus, it would be relevant to compare the effects of DMHIs that specifically target SAD with those that target anxiety disorders in general. To the best of the authors' knowledge, so far, no systematic review and meta-analysis has investigated the effect of DMHI on social anxiety in children and adolescents specifically.

Moderators in Treatment Outcome

DMHIs are not equally effective for all children, adolescents and young adults. The subsequent section will discuss potential moderators of treatment efficacy, which include control conditions, psychological principles of the interventions, age, adherence and the extent of support. Generally, the effect sizes are larger when DMHIs are compared with inactive control conditions (e.g., waiting list) than with active control conditions [57,58]. Most DMHIs developed are based on CBT principles [37] though when compared to DMHIs based on other psychological interventions (e.g., Cognitive bias modification (CBM), mindfulness) they do not result in better effects [59]. Evidence concerning age shows that older adolescents and young adults generally benefit more from DMHIs than children [46,60,61]. Similarly, adherence is supposed to be linked to symptom improvement, reflecting compliance with a predefined and recommended use and engagement with the intervention [62–64]. However, results regarding the association between adherence and outcome are more complex. Some studies point to a direct link between adherence and symptom improvement [65], some to an inverse relationship [66] and others identify only single aspects of adherence (e.g., module completion) to be relevant for symptom improvement [67]. Furthermore, studies indicate that adherence itself is, in turn, influenced by symptom severity, which may thereby affect symptom reduction [62]. Overall, adherence plays an important role in implementing DMHIs, and human support is a known factor in improving adherence [68,69]. This support can be provided in the form of guidance by professionals (i.e., psychologists and therapists) or by the social context of the young people (i.e., parents and teachers).

Objectives of this Systematic Review and Meta-Analysis

DMHIs are especially promising in young people as they are digital natives and generally seek help later and to a lesser extent than adults [38,70,71]. Thereby, the low-threshold and confidential nature of DMHIs may appeal especially to young people with SAD [30,72]. Thus, this systematic review and meta-analysis aims to evaluate the efficacy of DMHI in reducing social anxiety symptoms in children, adolescents and young adults.

Our primary research question is: Are DMHIs efficacious in reducing social anxiety symptoms in children, adolescents and young adults? Secondary research questions include moderator analyses: Is the between-group effect size of DMHIs moderated by control condition, age, the underlying psychological principle of the DMHI (i.e., CBT, other interventions), the intervention focus (SAD-specific or anxiety disorders in general), or human support (i.e., guided self-help vs. unguided self-

help and parental involvement vs. no involvement)? In addition to evaluating efficacy, we will systematically synthesize and assess adherence to the DMHIs, focusing on how adherence is defined, measured, and reported across studies.

Method

The study protocol for this review and meta-analysis was published on the International Prospective Register of Systematic Reviews PROSPERO, ID: CRD42023424181, on 15.08.2023 and conducted following the PRISMA guidelines.

Search Strategy and Study Selection

Search Strategy

Six electronic databases (APA PsycINFO, Embase, MEDLINE, PsyIndex, PubMed, Web of Science) were searched on 10 July 2024. The search was restricted to journal articles in the PsycInfo and Medline databases and randomised controlled trials in the Embase, Medline and Pubmed databases. Details for the full search string and all syntaxes are given in the multimedia appendix 1. Furthermore, the clinical trials database ClinicalTrials.gov was searched on 31 July 2024 for studies not identified in the electronic search, which yielded no additional studies. Previously published meta-analyses on DMHIs for anxiety disorders in children and adolescents were checked for studies not found with the database search.

Eligibility Criteria

Studies needed to meet the following inclusion criteria: 1) including young people with a mean age between 0 to <25 years [73,74], 2) evaluating a DMHI based on psychological principles that can be used remotely, 3) in a randomised controlled trial (RCT) comparing an experimental group to another active intervention, a waitlist, care-as-usual or treatment-as-usual condition, 4) reporting an outcome measure assessing social anxiety and 5) is published in a peer-reviewed journal in English or German. Study protocols, reviews, or meta-analyses were excluded.

The age range of under 25 years was selected to reflect current theories in child and adolescent psychiatry [75,76] and epidemiological evidence indicating the age of onset of SAD often occurs in this age range [6]. The rationale for including both prevention and treatment in the diagnostic spectrum is based on the understanding that social anxiety symptoms exist on a continuum, and the intervention techniques underlying both the prevention and treatment of SAD are similar [12,13,21,77–79].

Study Selection

Titles, abstracts and full texts were systematically reviewed. Author NW extracted all relevant information (title, abstract) for the study selection process to Excel. Independently, authors NW and AF first screened all titles and second all abstracts of the records identified through the searches. No specialised software was used. In case of discrepancies between the raters, the records were moved to the next screening round. Afterwards, NW and AF reviewed the full texts of the remaining records. If they disagreed about inclusion or exclusion, they discussed each study decision until consensus. If they did not reach an agreement, the final decision was taken together with SJS. Since the consensus method was followed, no inter-rater reliability was calculated.

Data Extraction

The data were extracted by two authors (NW and AF) independently into an Excel file as prepared in advance following the Cochrane Data Collection Form (Higgins et al., 2020). No automation tools were used. Information extracted from primary studies included study characteristics (i.e., country, ethical approval, trial preregistration, funding, study aims, study design, recruitment method, randomisation procedure, justification of sample size, dropouts, handling of missing values), sample

characteristics (i.e., informed consent procedure, eligibility criteria, sample size, sociodemographic information, clinical status), intervention characteristics (description of intervention, psychological principles, duration, amount of modules and delivery setting of intervention, extent of guidance, guidance provider and training thereof, adherence to and satisfaction with the intervention), control condition characteristics (i.e., waitlist, care-as-usual, treatment-as-usual, another active intervention, description and psychological principles thereof) and outcome data for all available measurement timepoints (i.e., assessment tool, sample size, unstandardized means, standard deviation (SD) and standard error, if SD was not available). If transdiagnostic anxiety measures were used and no outcome data for SAD were reported, corresponding authors were contacted. Authors who provided us with unpublished outcome data on SAD are indicated in Table 1.

Data Analysis

Data analyses followed the practical guide for meta-analyses in mental health research [80] and were analysed with the Comprehensive Meta-Analysis (CMA) version 4.0 software [81]. For each study, a standardised effect size was computed using Hedges' g and the 95% confidence intervals derived from means, standard deviations and sample sizes of the intervention and control groups. To account for potential heterogeneity, all analyses were computed with random effects models [80]. In 3-armed RCTs, we prioritised comparisons with the waitlist control group over another active control group as there were more studies with passive comparison conditions leading to a more homogenous comparison group [e.g., 82–85].

Heterogeneity was assessed through visual inspection of the forest plot, the Q-test and the calculation of the I^2 statistic. The Q-test assesses whether the observed effect sizes significantly differ from each other beyond chance. A significant Q-test indicates heterogeneity, but in small meta-analyses, there might be insufficient power to detect it [80]. I^2 values explain the proportion of heterogeneity, that is, the variability in treatment effect estimates due to real study differences and not due to chance, with 25% indicating low, 50% moderate and 75% high heterogeneity [86]. Additionally, we calculated Tau and Tau^2 . Tau measures the standard deviation of the true effects across studies, while Tau^2 is its square representing the estimated between-study variance. A significant Tau^2 suggests heterogeneity, supporting the use of random effects models [87].

As high heterogeneity was expected, we explored possible sources of heterogeneity in moderator analyses. Moderators were explored in subgroup analyses if the moderator was categorical (i.e., control condition, Risk of Bias, psychological principles, SAD-specific intervention, guidance, parental involvement, age group) or in a meta-regression analysis if the moderator was continuous (i.e., age, symptom severity). Subgroup analyses included the computation of average effect sizes and the Q-statistic as an indicator of a common effect size. A significant Q-statistic suggests heterogeneity in effect sizes within the subgroup.

Sensitivity analyses were performed to investigate the robustness of our findings. We conducted analyses excluding studies with the highest and lowest effect sizes to assess the potential influence of outliers on the meta-analytic results. This allowed us to evaluate the stability of the overall effect size when extreme values were omitted. In another sensitivity analysis, we only included studies with participants who all met diagnostic criteria for SAD to control for heterogeneous diagnostic samples. Moreover, in a subgroup analysis, studies were stratified based on their Risk of Bias ratings to investigate the influence of the methodological quality. In addition to these preregistered analyses, we conducted a sensitivity analysis by excluding studies with an age range over 25 years to test whether older participants altered the effects and we carried out a moderator analysis with age groups (children (>10 years), adolescents (11-17 years), young adults (<18 years)) as moderator.

Additionally, numbers needed to treat (NNT) were computed for all effect sizes using the *dmetar* package in R with the default Kraemer and Kupfer method [88]. NNT indicate the number of participants to be treated to achieve one case with reliable symptom change.

To address a possible publication bias, we visually inspected the funnel plot for symmetry and

conducted Begg and Mazumdar's test and Egger's test to quantify asymmetry. Using the approach by Duval and Tweedie, we assessed missing studies in the funnel plot to estimate the effect size after imputation [80,89–91].

Risk of Bias Assessment

Risk of bias was assessed by two reviewers (NW, AF) independently following the Risk of Bias Assessment (RoB2) procedure and using the RoB2 Excel template [92]. After the individual rating, discrepancies between ratings were discussed until consensus was reached. The RoB2 tool includes five domains for randomised trials: 1) Bias arising from the randomisation process, 2) bias due to deviations from intended interventions, 3) bias due to missing outcome data, 4) bias in the measurement of the outcome and 5) bias in the selection of the reported result. Risk of bias was rated separately for each domain based on whether it was likely to have influenced study outcomes. Studies categorised as low overall risk have low bias ratings in all five domains. Some concerns in at least one domain led to some concerns about the overall judgment of the study. High Risk of Bias was assigned to studies with at least one domain rated as high risk [80].

Results

Study Selection

The systematic literature search yielded 2149 records over all electronic databases. 474 records were removed before screening as they were duplicates over different databases. After the title screening, 1357 records, and after the abstract screening, 205 records, respectively, were excluded due to nonfulfillment of eligibility criteria. Therefore, 113 full texts were assessed for eligibility. Finally, 22 records were included in the systematic review and 21 in the meta-analysis. The study by Vigerland and colleagues [93] was not included in the meta-analytic calculations because there was insufficient data of social anxiety at post-assessment. For a complete overview, consider the flow diagram in Figure 1.

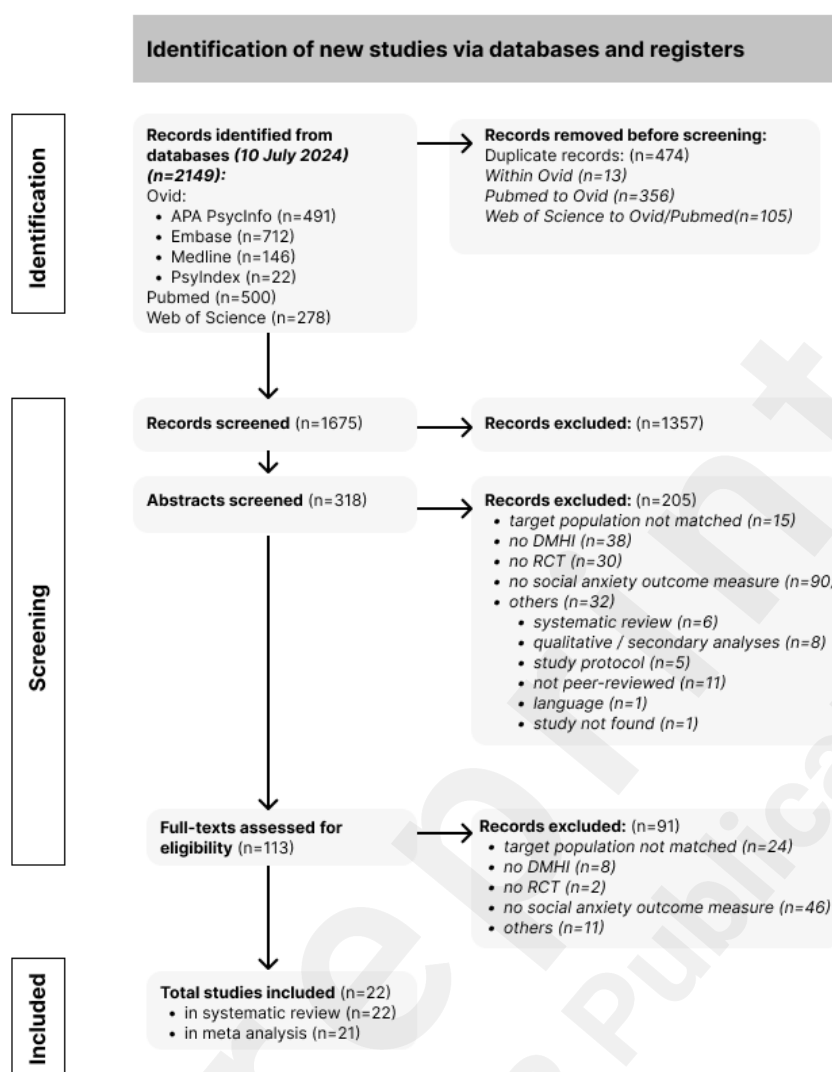


Figure 1. Flow diagram of identification and screening process.

Study Characteristics

The selected 22 papers were published between 2011 and 2024. The 22 trials were conducted in Australia ($k = 7$), USA ($k=4$), Sweden ($k=3$), UK ($k=2$), China ($k=2$), Spain ($k=1$), Iran ($k=1$), Canada ($k=1$) and the Netherlands ($k=1$). For a detailed overview of study characteristics, see Table 1.

Sample Characteristics

Participants were recruited via different strategies, mostly sought in self-selected sampling procedures for instance through posters and flyers, word of mouth, social media and online advertisements, postings on parental or health care providers, or media and newsletter articles [56,82,85,93–102]. The remaining RCTs recruited participants via convenience sampling (i.e., students research pool [103,104], school-based screening procedures [83,105–108], or an associated mental health clinic [109]. Few studies mentioned efforts to ensure a broad socioeconomic range either by choosing schools that cover multiple socioeconomic backgrounds [108] or by aligning the demographic profile of the study cohort to the country [56].

In total, 4196 participants were randomised across all RCTs. Their age spanned from 3 to 48 years, with means from 4.8 to 24.9 years. In all studies, more than half of the participants were female. A

comprehensive compilation of participant-related characteristics, including ethnic diversity and socioeconomic information on the final sample, can be found in Table 1.



Table 1.
Overview of included studies.

Study	Country	Participants							Study						Dropout
(1 st Author & Year)		Recruitment strategy	N Total, (I:C)	Age M(SD) ; range	Gender	% with SAD diagnosis	Ethnic diversity	Socio-economic information	Study design	Measurement timepoints	SAD outcome measure	Intervention name	Control condition	Control condition	% without post- assessme nt
(Rad et al., 2024)	Iran	School-based Screening	54 (18:18:18)	17 (1.16); 15-19	f:100.0% m: 0.0%	100%	Iranian: 100.0%	84.3% belonged to average economic group; 15.5% belonged to lower economic groups	RCT (3-arm)	Baseline Post:14 weeks FU: 3 months	SPIN	ICBT	Face-to-face CBT	Waitlist	24.3%
Espinosa et al., 2024	Spain	Self-selected sample: Referrals by school counsellors	58	14.9 (2.0); 12-18	f: 81.0% m: 19.0%	31%	Caucasian: 90%	Yearly family income: >10'000€: 23.2% 10'000 – 25'000€: 57.1% >25.'000: 19.6%	RCT (2-arm)	Baseline Post: FU: 3 months	RCADS-30 Social Phobia Subscale	AMtE	UP-A telehealth administered	N/A	16.7%
Mao et al. 2023	China	School-based Screening	30 (15:15)	15.1 (2.6); 14-17	f: 64.3% m: 35.7%	No clinician assessed diagnosis, all scored in top 10% of SAS-A	Not reported	Not reported	RCT (2-arm)	Baseline Post: 4 weeks	SAS-A	CBM-I-app	Waitlist	N/A	6.6%
(Hilt et al., 2023) ^a	USA	Self-selected sample • letters to guardians of 6-9th grade students at public schools • word-of-mouth • study posters in public spaces • online advertisements	152 (80:72)	13.7 (0.9); 12-15	f: 58.6% m: 41.5%	No clinician assessed diagnosis	White: 82.3% Black: 3.3% Asian: 2.0% Native: 0.7% Multiracial: 10.5% Other: 1.3% Non-Hispanic: 89.5% Hispanic: 10.5%	median household income 90'000-100'000\$ 9.21% recipients of government-assisted food program	RCT (2-arm)	Baseline Post: 3 weeks FU: 6, 12 weeks, 6 months	MASC-SA	App-based mindfulness exercises	Mood monitoring	N/A	0.00%
(Mueller & Cougle, 2023)	USA	Convenience sample: • Online advertisements • Student research pool	55 (28:27)	19.45 (2.1)	f: 85.2% m: 14.8%	100.0%	White: 77.7% Black: 12.1% Asian: 5.5% Latino*a: 22.2% Non-hispanic: 75.9%	Not reported	RCT (2-arm)	Baseline Post: 4 weeks FU: 1 month	SPIN	Building Closer Friendships	Waitlist	N/A	23.64%
(Leigh & Clark, 2022)	UK	School-based Screening	43 (22:21)	16.2 (1.1); 14-18	f:91.0% m: 9.0%	100.0%	48.8% from minority ethnic backgrounds	Not reported	RCT (2-arm)	Baseline Post: 14 weeks FU: 3, 6 months	LSASCA-SR	OSCA	Waitlist	N/A	4.65%
(Bautista et al., 2022)	USA	Self-selected sample: • Flyers on the	35 (20:15)	21.9 (4.8);	f: 71.4% m: 28.6%	No clinician	European American: 71.4%	Not reported	RCT (2-arm)	Baseline Post: 6 weeks	SIAS	Social Phobia	Waitlist	N/A	17.14%

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(Schniering et al., 2022) ^a	Australia	Clinical sample: Adolescents and their parents contacting the clinic for treatment were initially screened	91 (45:46)	14.3 (1.6); 12-17	f: 66.0% m: 34.0%	28.5%	Not reported	Not reported	RCT (2-arm)	Baseline Post: 8 weeks FU: 3 months (intervention group only)	SCASy	Chilled Plus Program	Waitlist	N/A	14.46%
(Nordh et al., 2021)	Sweden	Self-selected sample • Referrals from health care professionals • Advertisements at CAMHS clinics, newspapers + social media	103 (51:52)	14.1 (2.1); 10-17	f: 77.0% m: 23.0%	100.0%	Not reported	Not reported	RCT (2-arm)	Baseline Post: 10 weeks FU: 3 months	LSAS-C	ICBT for SAD	ISUPPORT: Psychoeducation about SAD + healthy habits	N/A	1.94%
(Stapinski et al., 2021)	Australia	Self-selected sample: • Media coverage • Social media postings	123 (62:61)	21.6 (2.2); 17-24	f: 67.0% m: 31.7% d: 0.8%	No clinician assessed diagnosis	82.1%: Australian-born 17.9%: of other origin	Not reported	RCT (2-arm)	Baseline: Post: 2 months FU: 6 months	Composite Score of SIAS & SPIN	Inroads	Psycho-education on alcohol	N/A	30.65%
(Wang et al., 2020)	China	Self-selected sample: • Advertisements on different internet platforms	randomised 210 (70:70:70); analysed: 104 (33:37:24)	24.9 (4.6); 18-45	f: 70.2% m: 29.8%	100.0%	Not reported	Not reported	RCT (3-arm)	Baseline Post: 8 weeks	SIAS	ICBT guided for SAD	ICBT self-help for SAD	Waitlist	50.48%
(Farrer et al., 2019)	Australia	Self-selected sample: • Social media postings School-based through newspaper, email	200	22.1 (4.9); 18<	f: 77.5% m: 17.0% d: 5.5%	No clinician assessed diagnosis	Caucasian/ European: 64.5% Asian/Indian: 28.0% Aboriginal/ Torres Strait Islander/Pacific Islander: 1.0% Latino/South American: 1.5% African: 1.0% Other: 4.0%	No financial stress: 28% Occasional financial stress: 42.5% Frequent financial stress: 21.0% Constant financial stress: 8.5%	RCT (2-arm)	Baseline Post: 6 weeks FU: 3 months	SOPHS	Uni Virtual Clinic	Waitlist	N/A	28.00%
(McCall et al., 2018)	Canada	Convenience sample: • Student research pool	101 (51:50)	21.9 (5.5) 17-46	f: 72.0% m: 28.0%	No clinician assessed diagnosis, Elevated score of Mini-SPIN and no prior SAD	White: 18.0% Asian: 62.0% Other: 20.0%	Not reported	RCT (2-arm)	Baseline Post: 4 months	SIAS	Overcome Social Anxiety	Waitlist	N/A	35.64%
https://preprints.jmir.org/preprint/67067															

JMIR Preprints						diagnosis									
(Spence et al., 2017 and Hearn et al., 2018) ^b	Australia	Self-selected sample: • via schools • parent groups • mental health professionals • guidance officers • media, Facebook	125 (47:48:30)	11.3 (2.7); 8-17	f: 60.0% m: 40.0%	100.0%	Country of Birth: Australia: 83.2% USA/Canada: 4.8% United Kingdom: 4.0% New Zealand: 2.4% Europe: 1.6% Africa: 1.6% Asia: 0.8% Other: 1.6% Indigenous Australian: 2.4%	High (>AUS\$100,001): 57.9% Low + Middle (<100,000): 42.0%	RCT (3-arm)	Baseline Post: 12 weeks FU: 6 months	SPAI-C	ICBT-SAD	ICBT- Generic	Waitlist	21.60%
(Morgan et al., 2017)	Australia	Self-selected sample: • Paid google + facebook ads • Postings on parenting + mental health websites • Flyers in preschool services	433 (215:218)	4.8 (1.0); 3-6	f:52.3% m: 47.7%	No clinician assessed diagnosis	Not reported	Financial difficulty: 17.1%	RCT (2-arm)	Baseline Post: 12 weeks FU: 24 weeks	PAS-R SAD subscale	Cool Kids for parents	Waitlist	N/A	20.79%
(Sanchez et al., 2017)	USA	Self-selected sample: • Social media postings • Postings on parenting list servers • Referrals by national school systems and child service providers	69 (33:36)	8.9 (1.2) 7-11	f: 40.5% m: 59.5%	No clinician assessed diagnosis	White/Non-White: I: 55.0%/45.0%, C:56.0%/44.0%, Hispanic/Non-Hispanic: I: 6.0%/94.0%, C: 6.0%/94.0%	Not reported	RCT (2-arm)	Baseline Post: 9 weeks	SASC-R	Adventures aboard the S.S. GRIN	Waitlist	N/A	Not indicated
(Calear, Christensen, et al., 2016)	Australia	School-based sample: Recruited schools from the Australian Capital Territory and South Australia	225 (127:98)	15.0 (1.08); 13-17	f: 70.6% m: 16.8%	No clinician assessed diagnosis	Not reported	Not reported	Cluster-RCT (2-arm)	Baseline Post: 6 weeks FU: 3 months	SAS-A	E-couch Anxiety and Worry program	Waitlist	N/A	I: 38.8% C: 27:6%
(Calear, Batterham, et al., 2016)	Australia	School-based sample of six participating headspace centres	1767	14.8 (1.0); 12-18	f: 62.8% m: 37.2%	No clinician assessed diagnosis	Not reported	Not reported	Cluster-RCT (3-arm)	Baseline Post: 6 weeks FU: 6, 12 months	SAS-A	e-GAD health services	e-GAD school	Waitlist	52.52%
(Vigerland, Ljótsson, et al., 2016)	Sweden	Self-selected sample: • Media advertisements	93 (46:47)	10.1 (1.7); 8-12	f: 55% m: 45%	10.0%	Not reported	Not reported	RCT (2-arm)	Baseline Post: 11 weeks FU: 3 months FU: 12 months ^c	SPAI-C/P	ICBT	Waitlist	N/A	96.77%
(Sportel et al., 2013)	Netherlands	N/A	240 (84:86:70)	14.1 (0.6); 12-15	f: 73.3% m: 26.7%	12.9% met SAD diagnosis; remaining elevated symptoms of social and/or test	Not reported	Not reported	Cluster-RCT (3-arm)	Baseline Post: 12 weeks FU: 6, 12 months	RCADS SAD Subscale	CBM-I	Group CBT	Waitlist	16.67%
https://preprints.jmir.org/preprint/67067															

JMIR Preprints						anxiety									
(Bowler et al., 2012)	UK	Self-selected sample: • posters • e-mail campaign	71 (24:25:22)	22.7 (5.9); 18-48	f: n=43 m: n=20	No clinician assessed diagnosis; elevated SPIN scores	primarily Caucasian	Not reported	RCT (3-arm)	Baseline Post: 2 weeks	SPIN	E-couch social anxiety program	CBM-I	Waitlist	11.27%
Tillfors (2011)	Sweden	Self-selected sample: • regional newspaper articles • school staff • advertisements in high school	19 (10:9)	16.5 (1.6); 15-21	f: n=17 m: n=2	100.0%	Not reported	Not reported	RCT (2-arm)	Baseline Post: 9 weeks FU: 1 year	SPSQ-C & LSAS-SR	CBT-based online-intervention	Waitlist	N/A	5.26%

Notes: Gender: f: = female; m = male; d = diverse;
Outcome Measures: SIAS = Social Interaction Anxiety Scale (Mattick & Clarke, 1998); SPIN = Social Phobia Inventory (Connor et al., 2000); LSAS-C = Liebowitz Social Anxiety Scale-Child version; LSASCA-SR = Liebowitz Social Anxiety Scale for Children and Adolescents- Self-report Version; (Masia-Warner et al., 2003); SCAS-C = Spence Children's Anxiety Scale; SCASy = Spence Children's Anxiety Scale, youth report; SCAS-P = Spence Children's Anxiety Scale, parent report (Spence, 1998); SAS-A = Social Anxiety Scale for Adolescents (La Greca & Lopez, 1998); SOPHS = Social Phobia Screener (Batterham et al., 2017); SPAI-C = Social Phobia and Anxiety Inventory for Children (Beidel et al., 2000); RCADS-SP = Revised Children's Anxiety and Depression Scale, Social Phobia Subscale (Chorpita et al., 2005); SPSQ-C = Social Phobia Screening Questionnaire for Children and adolescents (Gren-Landell et al., 2009)
^a We obtained unpublished, SAD-specific outcome data directly from the authors, which was not included in the original outcome paper.
^b:The data reported by Spence et al. (2017) were later used by Hearn et al. (2018); Spence compared three arms of the RCT with each other (transdiagnostic ICBT, SAD-specific ICBT and waitlist control), and Hearn compared both intervention groups to the waitlist group. Depending on the analysis, data were either drawn from Spence et al. (2017) or Hearn et al. (2018).
^c:12 months follow-up data were reported in a sperate publication (Vigerland et al., 2017).

Intervention Characteristics

For a detailed overview of intervention characteristics, consider Table 2. All interventions were delivered via computer, except two studies that investigated a smartphone application (app) [96,105]. In most studies, participants completed the interventions from home (or from anywhere) and only some interventions were delivered at a lab [82] or in a school setting [83] under the observation of a researcher or teacher, respectively. Except for the apps and the social skills game, all interventions were module-based. Participants either had access to all modules from the beginning or one module was activated per week.

Most interventions were based on CBT principles ($k=16$); some were CBM interventions ($k=3$) and some were based on mindfulness ($k=1$) or social skills training ($k=1$). The interventions either addressed anxiety symptoms or disorders in general or were specifically designed for SAD. SAD-specific interventions [56,85,98,101,103,104,106,108,110], particularly incorporated examples of fears and cognitions that are important in SAD (for example based on the Cognitive Model [12] or the cognitive-behavioral model [14]) or focused on different biases that play a role in SAD [82,84,105].

Adherence

Most commonly, adherence was operationalised as a mean number of completed modules per participant or percentage of participants that completed individual modules [56,82,84,93,94,100–102,104–107]. Leigh and Clark [108] reported the average time spent with the intervention and Nordh and colleagues [98] applied the “Internet Intervention Patient Adherence Scale” [111]. Only one study [100] analysed the effect of adherence on symptom reduction and found a dose effect with more completed modules resulting in a greater change from baseline to post treatment.

Guidance

The extent of guidance varied greatly over all studies. Nine studies were unguided self-help; in six studies participants had no contact [94,96,99,104,105,107] and three studies sent automated reminders or summary emails [84,97,103]. In two studies the presence of teachers or researchers ensured that participants were using the DMHI but they gave no feedback on the intervention progress [82,83]. Most programs individualised their human guidance in asynchronous weekly messages in the program, via email or as feedback on worksheets [56,85,93,98,100,101,108]. Some offered scheduled or on-request telephone support during the DMHI time [97,98,100,102,106,108,109]. One study offered weekly in-person meetings with a trained undergraduate student in the role of a peer coach [110]. Generally, guidance was provided by trained Psychology graduate students under regular supervision or by trained (CBT-)therapists. Some studies indicated time spent for guidance, e.g., [108] reported an overall mean of 398.67 minutes (6.64 h) spent directly communicating with participants. Other studies recorded the average effort per week and participant (15 to 30 minutes; [85,98]) or the average meeting time (5 to 35 minutes; [97,106,110]).

Table 2.
Characteristics of interventions.

RCT	Psychological principal	SAD-specific	Modules	Intervention duration	Adherence	Guidance format	Guidance amount	Parental involvement
(Rad et al., 2024)	Integrated CBT-based program based on CBT manual for anxious children (Kendal & Hedtke, 2006) and CBT for SAD (Hofmann & Otto, 2017)	Yes	10 modules	10 weeks	83.3% completed all 10 modules	Yes Weekly 5 min video call by a licenced therapist	Not reported	No
(Espinosa et al., 2024)	Transdiagnostic CBT program to support emotion regulation based on UP-A and previous Spanish program for adults	No	8 modules	8 weeks	≤ 1 modules completed: 100% Mean number of completed modules: 7.05 (SD=1.25)	(Yes) One phone call with adolescents/parents at the beginning and after intervention completion	Not reported	No
(Mao et al., 2023)	CBM including interpretation bias modification tasks	Yes	8 modules	4 weeks	93.3% completed entire training	No	Not reported	No
(Hilt et al., 2023)	Mood monitoring + Mindfulness exercises	No	Pax received mindfulness exercises after mood monitoring questions. Exercise length depended on availability of pax: 1, 5 or 10 minutes	3 weeks	Not reported	No	Not reported	No
(Mueller & Cogle, 2023)	Building closer friendships consisted of 3 components: 1) emotional writing, 2) social skills, 3) exposure exercises addressing individuals with SAD, specifically fear of intimacy	Yes	3 modules including 10 treatment components	4 weeks	≤ 1 treatment component completed: 100% Mean number of completed treatment components: 7.65 (SD=2.38)	No	Not reported	No
(Leigh & Clark, 2022)	OSCA: CBT-based program for SAD based on the Cognitive Model by Clark and Wells	Yes	8 core modules for weeks 1 and 2 Up to 16 additional modules to individualize treatment on specific problems and fears	14 weeks	Mean hours spent on OSCA: 26.14h (SD=11.32) Mean number of completed exercises: 25 (SD=10.75)	Yes Weekly 20 min phone calls and messages in the program by a clinical psychologist	Mean time spent in direct communication with participants: 398.67 minutes (SD =59.38)	No
(Bautista et al., 2022)	CBT-based program addressing SAD provided via this way up clinic.	Yes	6 modules	6 to 10 weeks	Not reported	Yes Weekly in-person meetings by undergraduate students who participated in 3 training sessions	Meeting time: M=11.73 minutes (SD=6.57)	No
(Schniering et al., 2022)	Chilled Plus is a CBT-based program addressing anxiety and depression based on the face-to-face Chilled program.	No	8 modules	8 weeks	Not reported	Yes Eight 30 min phone calls by therapists	Not reported	Yes
(Nordh et al., 2021)	CBT-based program addressing SAD based on the program by Nordh et al. (2017)	Yes	10 child modules 5 parent modules	10 weeks	Patient adherence scale rating: M=22.84 (SD=10.10)	Yes Three 30 minutes phone calls; asynchronous support by clinical psychologists	Therapists time per week M=28.85 (SD=16.79)	Yes
(Stamps et al., 2021)	Imroads/pre-GAM-67 program addressing anxiety and hazardous alcohol use	No	5 modules	8 weeks	≤ 1 module completed: 77.0% ≤ 3 modules completed: 51.0% All 5 modules completed: 39.0%	Yes Weekly emails; two 30 min phone calls or chat sessions	Not reported	No

						by clinical psychologists		
(Wang et al., 2020)	CBT-based treatment addressing SAD based on the program of Berger et al. (2009) (guided and unguided self-help)	Yes	5 modules	8 weeks	Not reported	Yes Weekly emails by psychology graduate students with training in CBT under supervision of a clinical psychologist	Mean time spent for guidance per participant per week= 15 minutes	No
					Not reported	No	Not reported	No
(Farrer et al., 2019)	Uni Virtual Clinic: CBT-based online mental health program	No	Multiple modules covering a topics students might be affected of (e.g., mood, anxiety, substance use, eating disorders and loneliness, homesickness, adjusting to university)	6 weeks	75.8%: accessed the program. 63.8% logged in weekly. 42.6% spent < than 5 minutes per visit	No	Not reported	No
(McCall et al., 2018)	Overcome Social Anxiety CBT-based addressing SAD	Yes	7 modules	4-6 months	Not reported	(No) Automated emails	Not reported	No
(Spence et al. 2017 and Hearn et al., 2018)	CBT-based generic anxiety program (BRAVE ONLINE)	No	10 modules of 60 min	12 weeks	Mean number of modules completed at postintervention: Children: 4.75 of 10 Their parents: 4.32 of 6 Adolescents: 4.00 of 10 Their parents: 3.18 of 5	Yes Weekly email feedback By trained psychologists supervised by an experienced clinical psychologist	Not reported	Yes
	CBT-based program addressing SAD that specifically focused on social anxiety and included social skills training and SAD specific cognitive elements based on the Clark and Wells (1995) model.	Yes						
(Morgan et al., 2017)	Cool little kids online: CBT-based program based on the cool little kids parenting group program addressing anxiety in children	No	8 modules	8 weeks	Not reported	Yes Automated summary emails; telephone support on request By psychologist	12 calls to 11 parents (Mean duration=35 minutes)	Yes
(Sanchez et al., 2017)	Adventures is an online game based on the social skills group intervention (DeRosier, 2004).	No	9 modules of 25 min	9 weeks	Not reported	No	Not reported	No
(Calear, Christensen, et al., 2016)	E-couch anxiety and worry program: CBT-based program addressing anxiety and worry	No	6 modules of 30-40 min	6 weeks	2 modules completed: 98.0% 4 modules completed: 68.0% All 6 modules completed: 45.0%	(No) But classroom teachers assisted with logins and supervised participants during time of program completion	Not reported	No
(Calear, Batterham, et al., 2016)	E-couch anxiety and worry program: CBT-based program mental health condition	No	6 modules of 30 to 40 min	6 weeks	Not reported	(No) Participants were supervised to complete the program and assisted if they had questions. By teachers and headspace education officers	Not reported	No
https://preprints.jmir.org/preprint/67067	E-couch anxiety and worry program: CBT-based program school condition	No	6 modules of 30 to 40 min	6 weeks	Not reported	(No) But classroom teachers assisted with logins and supervised participants during time of program	Not reported	No

(Vigerland, Ljótsson, et al., 2016)	CBT-based program addressing anxiety disorders	No	4 modules for children 7 modules for parents	10 weeks	Mean number of completed modules:9.7 (SD=1.8)	completion Yes online messages and written feedback on worksheets by psychologists/CBT-therapists	Not reported	Yes
(Sportel et al., 2013)	CBM including interpretation bias and attention bias modification tasks	Yes	20 sessions (40 minutes)	10 weeks	Mean number of completed sessions: 8.5 (SD=6.7)	(No) Automated E-mails to complete sessions	Not reported	No
(Bowler et al., 2012)	E-couch social anxiety program based on CBT principles CBM with 40 scenarios that relate to socially anxious persons per module	Yes	4 sessions	2 weeks	All participants completed all four sessions.	(No) Researcher ensured attendance and compliance with the program in the lab.	Not reported	No
(Tillfors et al., 2011)	CBT-based self-help manual addressing SAD based on a previously tested program for students	Yes	9 modules	9 weeks	Mean number of completed modules finished:2.9 (Range 1-6).	Yes feedback on the homework assignment by therapists	Not reported	No

Notes: CBT = Cognitive behavioral therapy; CBM = Cognitive bias modification; SAD = Social anxiety disorder; SD = standard deviation; M = mean

Risk of Bias

Risk of Bias assessment indicated that most studies showed some concerns (see Figure 2). A main concern was the fact that RCTs investigating psychological intervention do not facilitate adequate masking of participants to the treatment condition. Therefore, only studies with another active intervention that mimicked the initial intervention and in which participants were blinded to the condition (e.g., [98]) were rated with low risk in the dimension 4 *Measurement of the outcome*. Three studies were classified as high risk due to insufficient information on randomisation, no indication of analysis to estimate the effect of assignment to the intervention group, high missingness and no indication of any measures that account for that [85,94,99]. Insufficient information on the randomisation process (i.e., concealment of allocation sequence or assessment of baseline differences between groups) was reported in 39% (9/22) of the studies. In 30% (7/22) of the studies, there was no information on possible trial deviation or estimation of an effect of assignment to the intervention. Most studies (83%; 19/22) followed the intention-to-treat (ITT) approach and investigated their missing data. All but five studies (22%) reported their plan of analysis and the reporting of the result was transparent and complete. The Risk of Bias assessments of all studies are included in the supplementary material.

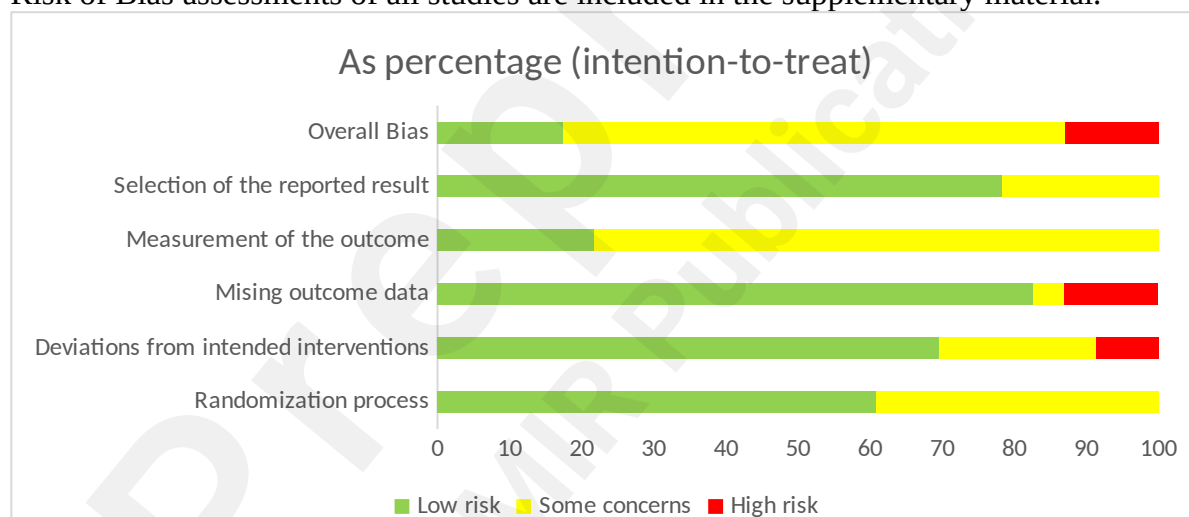


Figure 2. Overall Risk of Bias assessment displayed in percentages.

Effect of DMHIs on social anxiety symptoms

The random effects model yielded an overall effect size in favour of DMHI compared to any control condition at post-assessment of $g=0.508$ (95% CI: 0.308 to 0.707) (see Figure 3). Sensitivity analyses were performed a) excluding studies with the highest [108] and the lowest [83] effect size, b) excluding studies with an age range over 25 years [82,85,94,103] and c) including studies with the full sample being diagnosed with SAD [85, 95, 98, 101, 104, 106, 108]. Two additional analyses were performed for follow-up between-group effect sizes; one was computed with follow-up data collected 3 to 6 months and one with data collected 12 months after post-assessment. For a detailed overview, see Table 3.

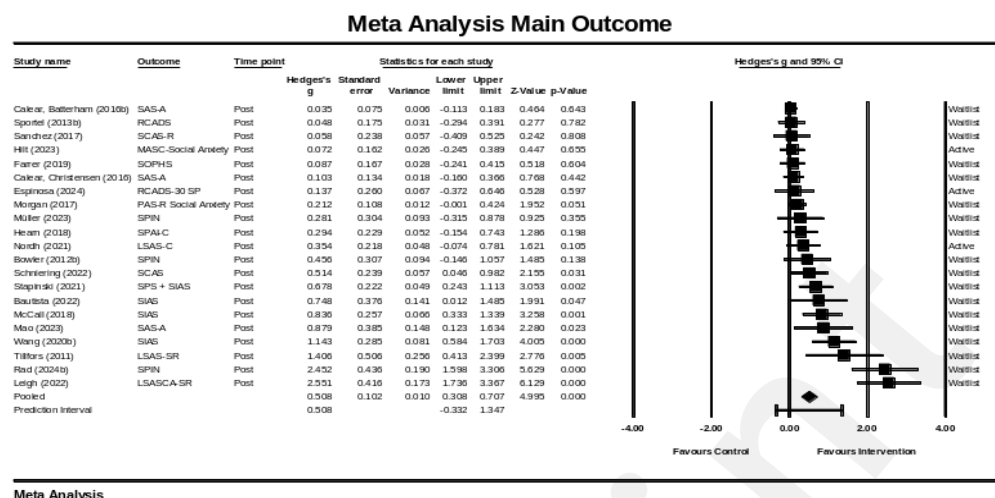


Figure 3. Forest plot of all studies included in the main outcome meta-analysis with the pooled effect size from the random effects model.

Subgroup analyses

In subgroup analyses, the moderating effect of the control condition, Risk of Bias, age group, underlying psychological principles, the focus of the intervention on SAD or transdiagnostic anxiety symptoms, guidance and parental involvement was investigated. Except for the subgroup analysis comparing the effect of control conditions, both active and inactive conditions were included in all analyses. The effect size was significantly greater if the DMHI was compared to a waitlist ($g=0.576$, 95% CI: 0.343 to 0.809) than to another active intervention ($g=0.101$, 95% CI: 0.021 to 0.261). Furthermore, DMHIs based on CBT ($g=0.610$, 95% CI: 0.361 to 0.859) were significantly more effective than DMHIs based on other therapeutic interventions ($g=0.176$, 95% CI: -0.052 to 0.405) (i.e., CBM). DMHIs including therapeutic elements specifically addressing SAD ($g=0.878$, 95% CI: 0.469 to 1.278), were significantly more effective than DMHIs addressing transdiagnostic anxiety symptoms ($g=0.195$, 95% CI: 0.040 to 0.277). Guided self-help ($g=0.825$, 95% CI: 0.425 to 1.224) was more efficacious than unguided self-help ($g=0.271$, 95% CI: 0.083 to 0.458). Age groups (i.e., children: $g=0.185$, 95% CI: -0.008 to 0.379, adolescents: $g=0.575$, 95% CI: 0.270 to 0.880, young adults: $g=0.578$, 95% CI: 0.276 to 0.879) differed significantly with children having the smallest and a non-significant effect. No significant difference was found for DMHIs with ($g=0.281$, 95% CI: 0.117 to 0.445) or without ($g=0.602$, 95% CI: 0.242 to 0.962) parental involvement. The computed effect sizes for different levels of Risk of Bias were not significantly different from each other (i.e., low risk: $g=0.201$, 95% CI: -0.029 to 0.431; some concerns: $g=0.622$, 95% CI: 0.359 to 0.885; high risk: $g=0.399$, 95% CI: -0.213 to 1.011); however, while the prediction interval for the low and some-risk subgroup analyses were small, the one in the high-risk bias subgroup analysis spanned from 6.9 to 7.7. Consider Table 3 for a complete overview and the multimedia appendix (MA. 3) for all forest plots.

Table 3.
Effect sizes of DMHIs compared with control groups (random effects model at post-assessment if not otherwise indicated).

	N _{co}	Effect size		Test of null		Prediction interval		Between-study		Heterogeneity statistics				NNT
		Hedges' g	95% CI	Z (2-tail)	P	Lower Limit	Upper Limit	Tau	TauSq	Q-value	df (Q)	P-value	I ²	
Overall effect	21	0.508 (0.104)	0.308 to 0.707	4.995	>.001	-0.332	1.347	0.388	0.151	100.164	20	>.001	80.033	3.6
Sensitivity Analyses														
Highest effect size removed	20	0.414 (0.088)	0.242 to 0.587	4.705	>.001	-0.262	1.091	0.310	0.096	69.414	19	>.001	72.628	4.3
Lowest effect size removed	20	0.551 (0.111)	0.333 to 0.769	4.963	>.001	-0.358	1.461	0.418	0.175	88.354	19	>.001	78.496	3.3
3 to 6 months Follow-up	12	0.378 (0.110)	0.162 to 0.594	3.427	>.001	-0.366	1.122	0.315	0.099	50.871	11	>.001	78.376	4.8
12 months Follow-up	2	0.064 (0.086)	-0.105 to 0.233	0.748	0.455			0.000	0.000	0.168	1	0.682	0.000	27.7
Adults >25 omitted	17	0.485 (0.113)	0.264 to 0.706	4.306	<.001	-0.373	1.343	0.387	0.149	83.418	16	>.001	80.820	3.7
Only 100% SAD sample	7	1.149 (0.331)	0.510 to 1.798	3.473	0.001	-1.084	3.383	0.803	0.646	46.931	6	>.001	87.215	1.7
Subgroup Analyses														
Study characteristics														
Control condition										10.561	1	0.001		
Waitlist	18	0.576 (0.119)	0.343 to 0.809	4.842	<.001	-0.363	1.515	0.427	0.182	98.592	17	>.001	82.757	3.2
Active	9	0.141 (0.061)	0.021 to 0.261	2.310	0.021			0.000	0.000	4.068	8	0.851	0.000	12.6
Risk of Bias score										1.554	2	0.460		
Low risk	3	0.201 (0.117)	-0.029 to 0.431	1.715	0.086			0.000	0.000	1.417	2	0.493	0.000	8.9
Some concerns	15	0.622 (0.134)	0.359 to 0.885	4.633	<.001	-0.371	1.614	0.439	0.193	87.152	14	0.000	83.936	2.9
High risk	3	0.399 (0.312)	-0.213 to 1.011	1.277	0.202	-6.972	7.770	0.489	0.239	11.344	2	0.003	82.370	4.5
Age group										7.042	2	0.030		
Children (<10 yrs)	2	0.185 (0.099)	-0.008 to 0.379	1.877	0.060			0.000	0.000	0.346	1	0.556	0.000	9.6
Adolescents (11-17yrs)	12	0.575 (0.156)	0.270 to 0.880	3.697	<.001	-0.521	1.672	0.467	0.218	76.542	11	>.001	85.629	3.2
Young adults (>18 yrs)	7	0.578 (0.154)	0.276 to 0.879	3.754	<.001	-0.301	1.457	0.305	0.093	14.484	6	0.025	58.576	3.2
Intervention characteristics														
Psychological principal										6.302	1	0.012		
CBT	16	0.610 (0.127)	0.361 to 0.859	4.800	<.001	-0.361	1.580	0.434	0.189	93.551	15	0.000	83.966	3.0
Others	5	0.176 (0.117)	-0.052 to 0.405	1.512	0.729	-0.376	0.729	0.129	0.017	5.285	4	0.259	24.307	10.1
SAD-specific intervention										11.034	1	0.001		
Yes	12	0.878 (0.206)	0.469 to 1.278	4.231	<.001	-0.615	2.362	0.635	0.403	63.447	11	0.000	82.663	2.2
No (transdiagnostic)	9	0.195 (0.061)	0.040 to 0.277	2.622	0.009	-0.104	0.422	0.093	0.093	11.150	8	0.193	28.250	9.1
Guidance										6.065	1	0.014		
No	11	0.271 (0.096)	0.083 to 0.458	2.834	0.005	-0.301	0.842	0.234	0.055	27.308	10	0.002	63.381	6.6
Yes	10	0.825 (0.204)	0.425 to 1.224	4.049	0.000	-0.567	2.216	0.568	0.323	58.760	9	<.001	84.683	2.3

Parental involvement														
Yes	4	0.281 (0.084)	0.117 to 0.445	3.355	0.001			0.000	0.000	0.670	1	0.413		
No	10	0.602 (0.184)	0.242 to 0.962	3.275	0.001	-0.644	1.847	0.508	0.258	1.476	3	0.688	0.000	6.4
										74.031	9	<.001	87.843	3.0

Notes: N_{co} = Number of comparisons (i.e., number of primary studies that went into the meta-analytical evaluation and subgroup analyses).

Meta-regression of age and symptom severity

Age and baseline symptom severity of social anxiety were introduced as a moderator in the main analysis. The meta-regression indicated that age had no significant effect on the efficacy of DMHIs on social anxiety ($\beta=.038$, 95%: -0.003 to 0.079, $p=.067$). Higher baseline symptom severity increased the effect size significantly ($\beta=.014$, 95%: 0.005 to 0.023, $p=.002$).

Publication Bias

Visual inspection of the funnel plot suggested the presence of a publication bias as the standard errors were not equally scattered around the overall mean (see Figure 4). Both the Begg and Mazumdar's test and the Egger's test of the intercept were significant (both $p>.001$), supporting an underlying publication bias. When adjusting for missing studies using the Duval and Tweedie trim and fill procedure in the random effects model, the effect size resulted in $g=0.506$ (95% CI= 0.308 to 0.707).

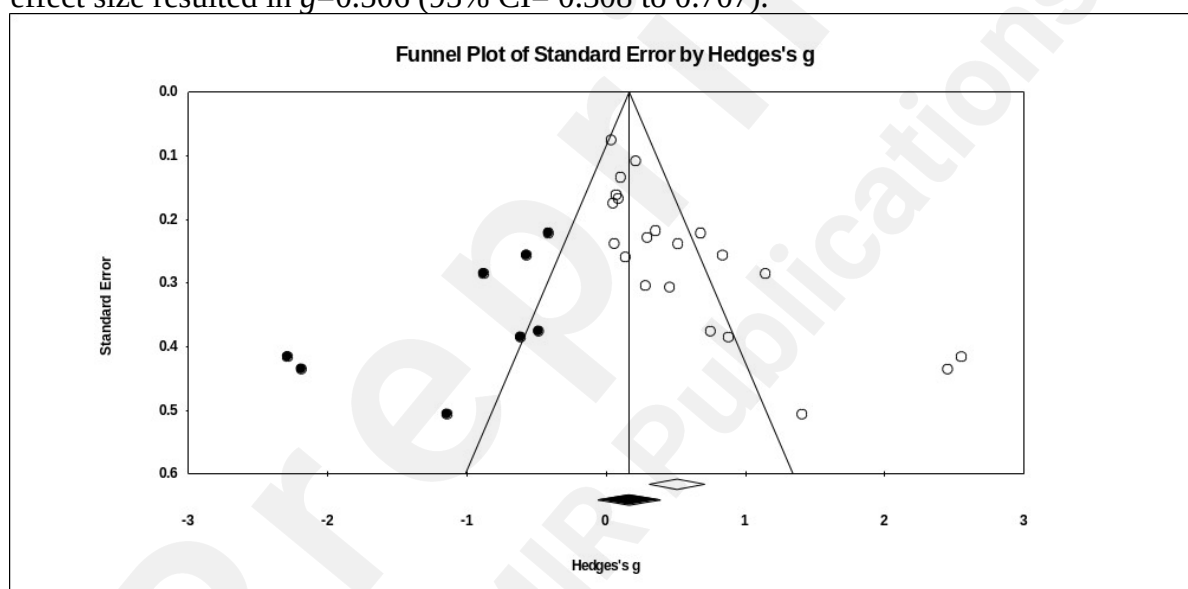


Figure 4. Funnel plot of all studies included in the meta-analysis and corrected mean effect (black diamond) and imputed effect sizes (black dots).

Discussion

This meta-analysis evaluated the efficacy of DMHIs on SAD in young people. Across 21 RCTs and 4196 participants, we found a small to medium pooled effect size in favour of DMHIs compared to both inactive and active control conditions assessed after the intervention. The effect size remained after taking the publication bias into account. Notably, the treatment gains decreased until 3 to 6 months follow-up and were no longer significant at 12 months follow-up. Subgroup analyses revealed that DMHIs based on CBT, including SAD-specific elements and support by human guidance carried out in adolescents and young adults resulted in higher pooled effect sizes compared to other psychological interventions, transdiagnostic interventions for several anxiety disorders and unguided self-help programmes in children. Mean age and parental involvement had no significant effect on the outcome.

Overall Effects in comparison with other meta-analyses

Compared to other meta-analyses of DMHIs addressing anxiety disorders, the overall effect size found in this meta-analysis is smaller. In a sample with the same age span but mixed anxiety disorders, a meta-analysis found a large pooled effect size ($g=0.68$) [45]. Even in children and adolescents at risk for an anxiety disorder (i.e., subthreshold of diagnosis), where smaller effect sizes would be expected, a large pooled effect size ($SMD=0.77$) was found [44]. Possible reasons for the smaller effect compared to previous meta-analyses could be the large heterogeneity between the included studies regarding the interventions, the samples and the study designs. Another reason could be the fact that despite the focus of this meta-analysis on SAD samples the primary studies included a broad range of mental health problems or disorders (e.g., other anxiety disorders, depression). Consequently, study participants did not necessarily suffer from social anxiety. In other meta-analyses, either a measure assessing various anxiety disorders was defined as the primary outcome, or the interventions and samples were primarily geared towards SAD. The sensitivity analysis in this meta-analysis showed that when only studies with participants with a SAD diagnosis were included, the calculated effect size was large ($g=1.15$). Meta-analytic evidence of face-to-face settings for children and adolescents diagnosed with SAD only also yielded a larger effect ($g=0.71$) that even increased until follow-up [22]. Similar effect sizes [112] were found in ICBTs for adults with SAD ($g=0.55$ [112]; $g=0.76$ [113]). In line with this, the moderator analyses showed that there are, in some cases, indeed large differences between the effect sizes of the subgroups. These differences are discussed in the following.

Psychological Principles of Effective DMHIs

The superiority of CBT-based treatments over other treatments, such as ICBM or mindfulness exercises is also reflected in other studies [58,114]. However, most treatments for young people with SAD are based on CBT and the evidence for other treatments is still limited [115]. Interestingly, SAD-specific treatments, including CBT and CBM, outperformed treatments addressing anxiety in general. This finding contradicts RCTs directly comparing two treatments (transdiagnostic anxiety vs. SAD-specific, i.e., [55,56]) but aligns with other evidence suggesting adolescents with SAD have poorer chances compared to other anxiety disorders [53,54,116]. This finding is quite striking because, DMHIs addressing SAD in adults have an extensive evidence base with RCTs demonstrating efficacious interventions based on CBT, psychodynamic therapy, and acceptance-based interventions [39] and meta-analytic evidence supporting ICBT to be equally efficacious compared to face-to-face CBT [42]. In young people, the question remains as to which treatment components have a generic effect in anxiety disorders and which would improve the treatment effect in SAD and thus indicate a disorder-specific treatment. Treatment components specific to SAD are theoretically discussed in aetiological and maintenance models of SAD [12–14]. So far, there is limited empirical evidence of whether these components are specifically important for young people with SAD [21]. It would be valuable to test both transdiagnostic anxiety and SAD-specific components individually in interventions. Compared to RCTs so far, study designs such as factorial or leapfrog designs would be more suitable for obtaining information on the efficacy of individual components and the optimal compilation of those components to an intervention [117–120].

Supporting Factors in DMHI Implementation for Young People

Similar to previous evidence, moderator analyses supported the efficacy of guided DMHIs but not unguided DMHIs [59]. Notably, there was great variety in the extent of human guidance: Some studies provided participants only with summaries of exercises via chat or email, while others phoned them regularly. The amount of support was quantified only in a minority of the trials. Furthermore, while some guidance formats reflected strategies to improve engagement with the intervention, the guidance of other trials could be viewed as an additional treatment in a format of blended therapy. To investigate how to optimise the amount, the content or the delivery setting of guidance for young people, future papers should be more detailed in their rapport of the guidance format and purpose and may even compare different ways of supporting young people in their studies [e.g., 121]. Possible results could streamline future implementation research and support the use of and continuous engagement with digital approaches.

Another aspect that varied considerably across studies was parental involvement. In one study, the intervention was only directed at parents [97], while others included specific parent information or modules that complemented the children's modules [56,93,98,109], or only addressed the young people. Research to date shows an inconclusive picture of how parents can be best involved in treatments to increase their overall effectiveness compared to child/adolescent only interventions. [122]. Comparable to our results, other meta-analyses also found no additional effect of parental involvement [123–125]. Especially in adolescence and young adulthood, the social context relevant to anxiety-inducing situations lies primarily outside the parental home (e.g., at school, with peers). Therefore, the involvement of parents might be less relevant for symptom reduction. Nevertheless, for young people parents play an important role in accessing mental health services [126,127] and could therefore be a crucial supporting factor in their children's adherence to the intervention [128].

Adherence has been defined and reported, if at all, in great variety in the included studies. Most commonly, completed modules or time spent in the DMHI were reported, but rarely, their effect on treatment outcome. Although these indices give some information on DMHI usage, they do not provide any information about reasons for non-adherence. Previously, some efforts have been made to find explanations for DMHI or treatment non-usage. One explanation given in adult samples is the *good enough effect*; that is, researchers explained the discontinuation of the DMHI in that the participants had reached their set goals [129]. This type of participant was also found in a qualitative examination of adolescents who stopped treatment for depression. Thereby, two additional types were also identified: the *dissatisfied*, those who did not find the intervention helpful and the *troubled*, those who reported a lack of stability in their lives to engage in an intervention [130]. Future research should report a minimum of adherence parameters (e.g., completed modules, time spent in the programme) and might even add reasons for non-adherence. Furthermore, it would be important to report the analysis of whether adherence influenced the outcome within the publication reporting the main results of the RCT. With this information, studies could be better compared and the influence of adherence on the outcome could be assessed meta-analytically [131].

Moderators of Treatment Efficacy

Regarding potential moderators, age did not emerge as a moderator of treatment efficacy in the meta-regression analysis, but subgroup analysis reported significantly higher effect sizes for adolescents and young adults compared to children. Likewise, other meta-

analyses have indicated that older adolescents profit more from DMHIs than children [37]. However, the small number of studies including child samples in this meta-analysis could also be a reason for this result. Furthermore, even though the influence of age was taken into account in this meta-analysis - using meta-regression (mean age), subgroup analyses (age groups), and by excluding studies with an age range over 25, many of the primary studies included wide age ranges and did not report results for different age groups. This highlights the need to replicate the results in future studies.

Symptom severity at the beginning of the intervention moderated treatment outcome with a small effect. In other studies, larger treatment effects were also found for adults with higher symptom severity [132]. The evidence in childhood and adolescence is mixed; some meta-analyses or studies found a significant effect while others did not [133]. This result, therefore, is in line with the overall mixed picture.

Designing and implementing DMHIs in the Future

This review included DMHIs that were developed primarily in high-income countries and provided in English, Swedish, Chinese, Persian, Spanish, or Dutch. Furthermore, few undertook efforts to match the socio-economic composition of the study with that of the home country [56,108]. A recurring argument advancing DMHIs is that they offer help at low cost, low threshold and high accessibility to young people from non-urban and lower financial sectors [38,134,135]. However, this presumption can only be realised if measures are taken to design DMHIs for diverse target groups, evaluate them in different demographic groups and support their implementation, particularly in those groups that otherwise have little access to mental health care. To achieve this, young people need to be actively involved in designing interventions and validation studies, recruiting participants for trials and communicating about and implementing digital self-help options [115,136].

Limitations and Implications for Future Research

This review and meta-analysis have some limitations and the results should be considered thoughtfully. Firstly, the meta-analysis and included subgroup analyses are based on a small number of studies and results should be interpreted cautiously. The limited number of studies precluded an examination of the efficacy of specific intervention components (e.g., psychoeducation, exposure therapy, social skills training, CBM or relaxation and mindfulness exercises) or for diagnostic categories (subclinical vs. clinical). Most studies included participants with elevated scores in social anxiety screenings based on self-reports and few of the identified prevention studies made diagnostic efforts to distinguish youths with subclinical social anxiety symptoms from those with a previous or current SAD diagnosis. Thus, prevention and treatment studies could not be categorised reliably to investigate potential differential treatment effects in subgroup analyses. Moreover, only two studies provided long-term follow-up data, resulting in limited information on the sustainability of treatment effects. Next, this meta-analysis encompassed a broad age range, spanning distinct developmental stages and diverse target groups. For instance, while adolescents are typically the primary end-users of digital interventions, in the case of preschool-aged children parents or legal guardians generally serve as the end-users, given that young children often lack the cognitive and emotional capacities to engage with such interventions independently. To explore potential age-related effects, subgroup analyses were conducted across three age categories (i.e., children, adolescents, young

adults). However, these analyses were limited by the small number of primary studies within each subgroup and did not account for developmental differences within or between age groups. Given the substantial developmental variability that exists within these broad categories, future research should examine potential moderator effects using more granular age distinctions and developmental profiles, particularly in terms of cognitive, emotional, and social competencies.

Furthermore, factors influencing the efficacy, such as adherence, were inadequately reported in primary studies, preventing their incorporation into the meta-analysis, for instance, in a meta-regression. Lastly, the methodological quality of the studies varied in terms of transparency in randomisation, masking procedures and consideration of appropriate statistical procedures.

Future intervention studies should, therefore, be methodologically more rigorous, include thorough diagnostic procedures, provide long-term follow-up data and report on potential moderators such as adherence, parental involvement, guidance and participant characteristics in more detail. Moreover, more details on how access to the modules was managed would be interesting as it may have an impact on the adherence of the participants as well as on the efficacy of the intervention. Furthermore, future studies should aim to target a socioeconomically diverse study population and report on demographic information that may influence the availability and efficacy of treatments (e.g., gender, age, country, urbanicity, socioeconomic status, culture and minoritised status, [137]). This additional information could inform future meta-analyses or, preferably, meta-analyses of individual participant data that don't rely on summary statistics to investigate subgroup effects.

Conclusions

In conclusion, while further research is necessary to provide conclusive evidence on the efficacy of DMHIs for young people, our meta-analysis and review shows their potential to alleviate SAD. The preliminary findings underscore the importance of future research efforts aimed at clarifying the specific treatment components that render DMHIs effective, optimising human guidance strategies and individualising interventions to enhance therapeutic outcomes. Moreover, our study highlights inclusivity in DMHIs and research design, emphasising the need for future studies to encompass participants from all socio-economic backgrounds. By addressing these considerations, DMHIs hold promise in bridging the care gap and democratising access to psychotherapeutic interventions for all young people, thus fostering mental health equity in the future.

Registration and protocol

The study protocol for this review and meta-analysis was published on the International Prospective Register of Systematic Reviews PROSPERO, ID: CRD42023424181, on 15.08.2023.

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Conflicts of Interest

No authors declare competing interests.

Author Contributions

Noemi Walder: Conceptualization, Methodology, Formal Analysis, Investigation, Writing – Original Draft, Project Administration; Alessja Frey: Formal Analysis, Investigation, Writing – Reviewing & Editing; Thomas Berger: Conceptualization, Methodology, Writing – Reviewing & Editing; Stefanie J Schmidt: Conceptualization, Methodology, Writing – Reviewing & Editing, Supervision

Abbreviations

CBM = Cognitive bias modification
CMA = Comprehensive Meta-Analysis
CBT = Cognitive behavioral therapy
DMHI = Digital mental health intervention
ICBT = Internet-based cognitive behavioral therapy
NNT = Number needed to treat
RCT = Randomised controlled trial
PROSPERO = Prospective Register of Systematic Reviews
PRISMA = Preferred Reporting Items for Systematic reviews and Meta-Analyses
ROB2 = Risk of bias assessment tool 2
SAD = Social anxiety disorder
SD = Standard deviation
SMD = Standardized mean deviation

Multimedia Appendix 1

Search Strings used for the systematic literature review.

Multimedia Appendix 2

Risk of Bias assessment of all studies over all five domains.

Multimedia Appendix 3

Forest plots for all meta-analyses.

Multimedia Appendix 4
Prisma Checklist.



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Supplementary Files

Multimedia Appendixes

Search Strings used for the systematic literature review.

URL: <http://asset.jmir.pub/assets/0b2ba5e9feaeade5c2d20c82c1469d03.docx>

Risk of Bias assessment of all studies over all five domains.

URL: <http://asset.jmir.pub/assets/42b2f9840dbf289c2be19299b51fe98b.docx>

Forrest plots for all meta-analyses.

URL: <http://asset.jmir.pub/assets/9cc8d85031903d52c9eca4ac487f0dc0.pptx>

Prisma Checklist.

URL: <http://asset.jmir.pub/assets/f3973bd720df90deb74735153714adba.docx>